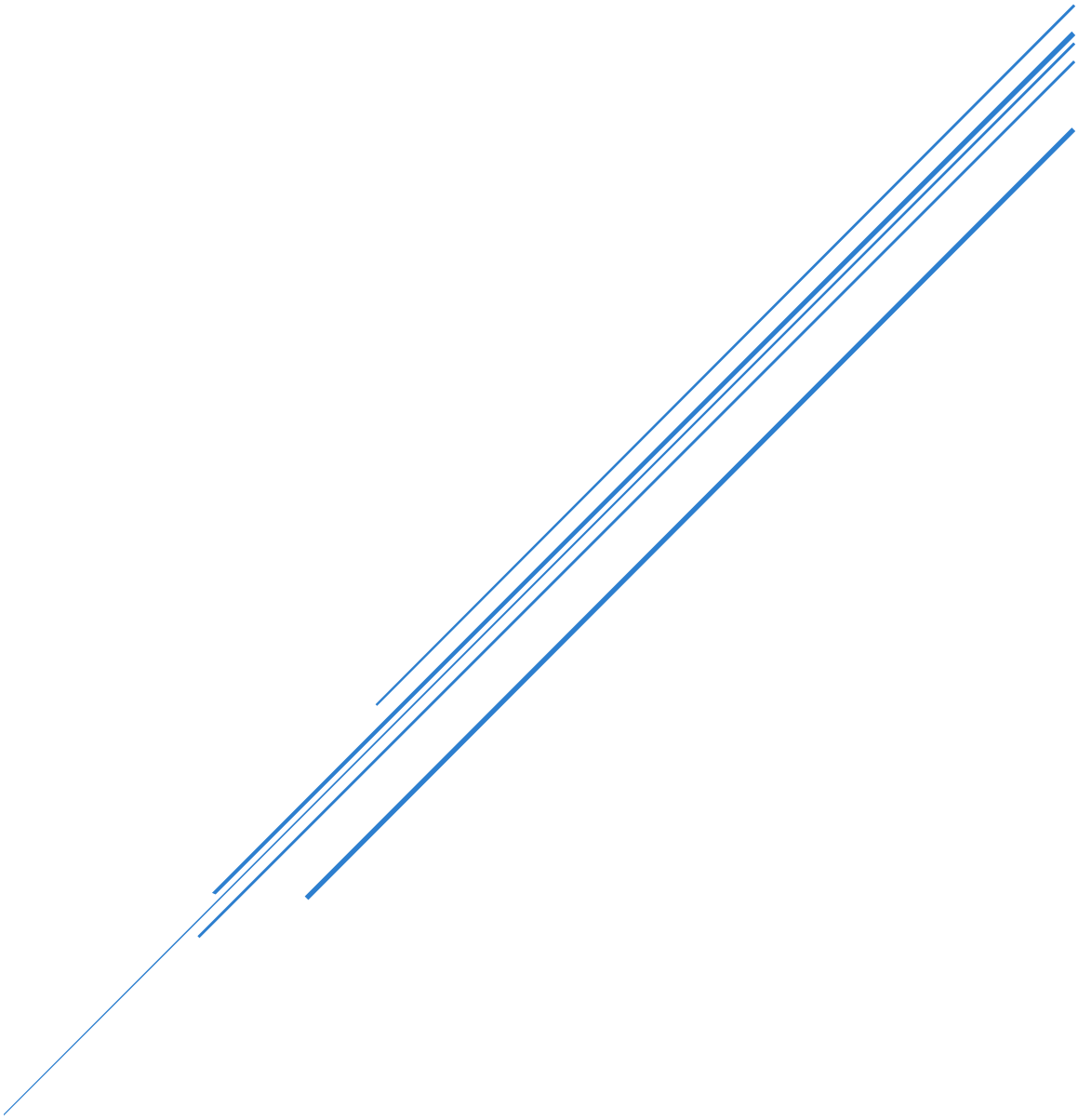
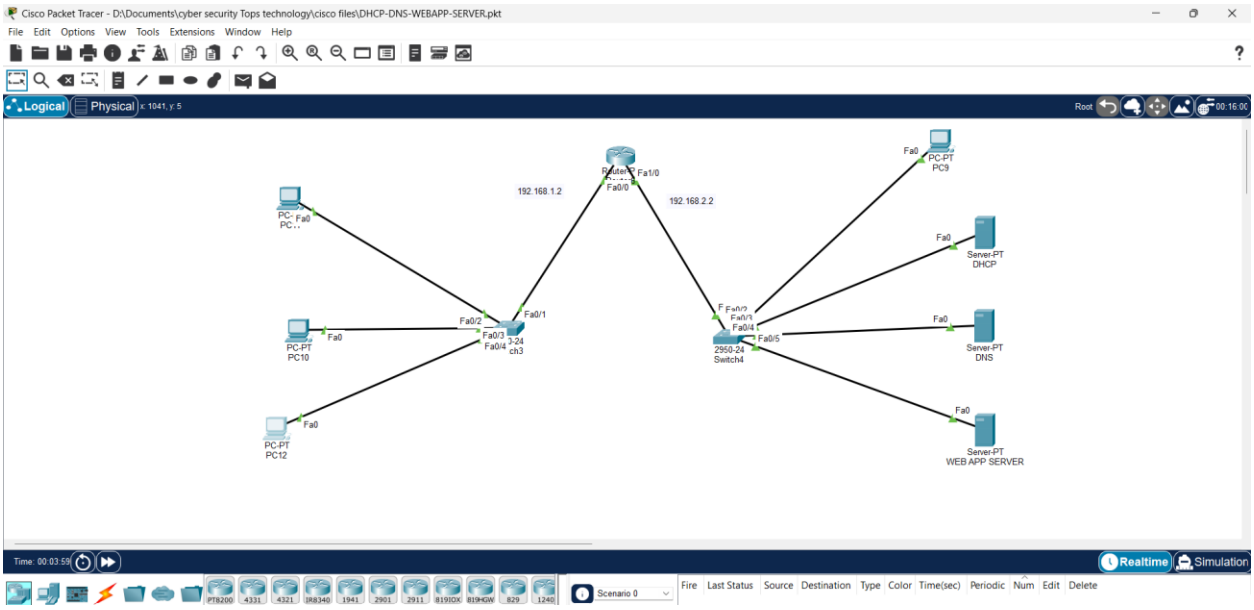
DAY - 2 DHCP CONFIGURATION

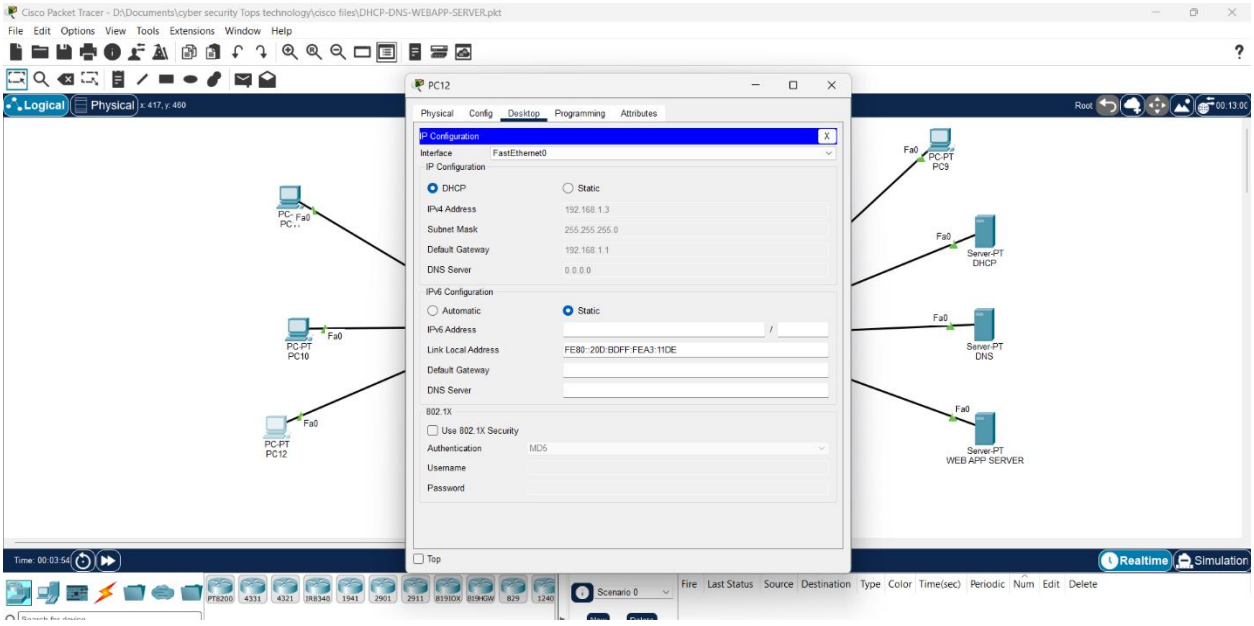
Assignment - 10



1. **In Cisco Packet Tracer, set up a simple network with one DHCP server and two client PCs. Configure the DHCP server to assign IP addresses automatically, then verify on each PC that it receives a different IP address from the DHCP pool.**

➢





I created a network in Cisco Packet Tracer using one DHCP server and two client PCs connected through a switch. The DHCP server was configured to automatically assign IP addresses to the client PCs.

Steps Performed:

1. Connected the DHCP server and two PCs to a network switch.
2. Configured the DHCP server with a static IP address.
3. Enabled the DHCP service on the server.
4. Created a DHCP pool with the network address, subnet mask, default gateway, and starting IP address.
5. Opened each PC's Desktop → IP Configuration and selected DHCP.
6. Verified that both PCs received different IP addresses automatically.

IP Address Verification:

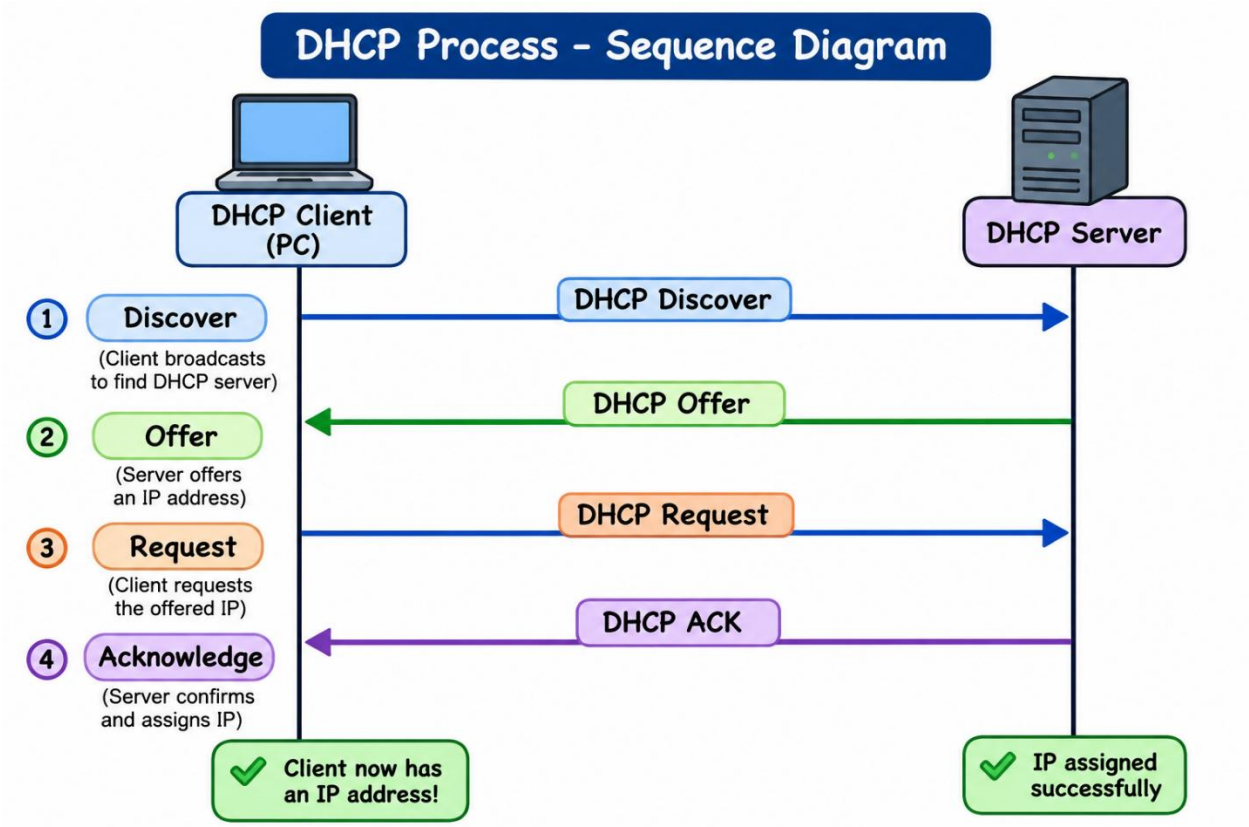
Device	IP Address	Subnet Mask
PC1	192.168.1.2	255.255.255.0
PC2	192.168.1.3	255.255.255.0

Result:

Both PCs successfully received different IP addresses from the DHCP server automatically. This confirms that DHCP is working correctly and manual IP configuration is not required for client PCs.

2. **Draw a sequence diagram (hand-drawn or digital) that clearly shows each step of the DHCP process: Discover, Offer, Request, and Acknowledge, labeling the sender and receiver at each stage.**
   Hint: Use arrows to show the direction of each message between client and server.

➢



DHCP Process – Sequence Diagram

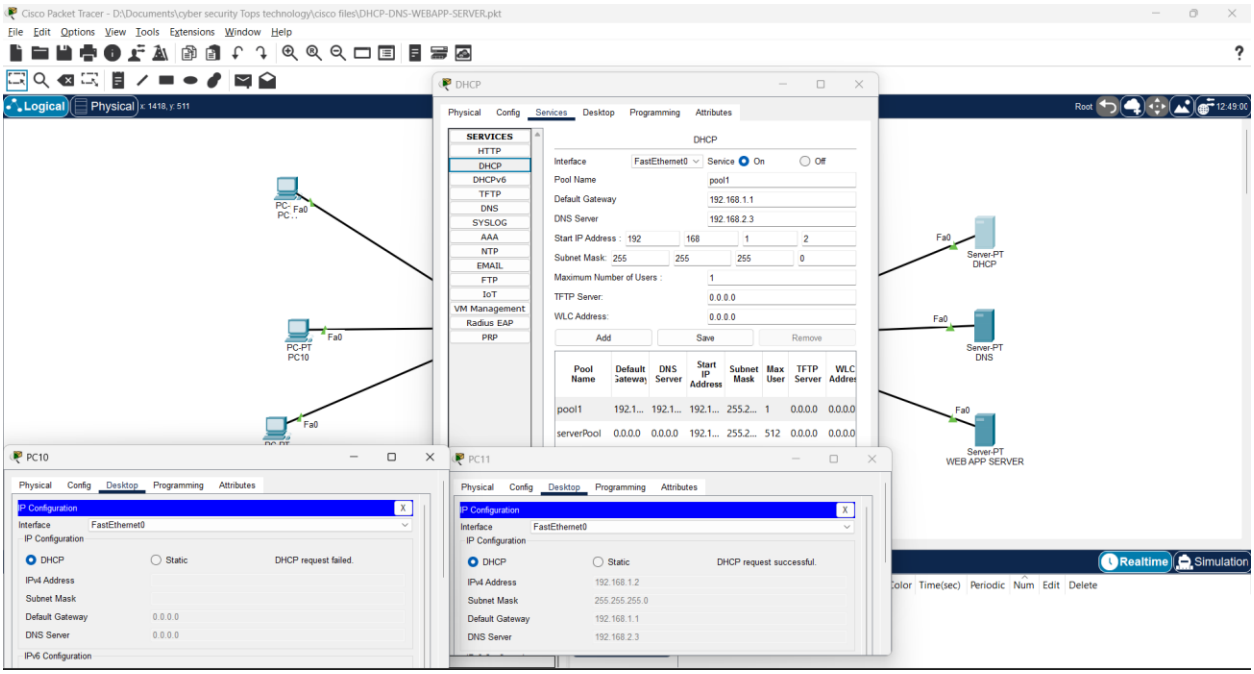
The DHCP process consists of four steps, known as DORA:

1. Discover: The client sends a broadcast message to find an available DHCP server.
2. Offer: The DHCP server offers an available IP address to the client.
3. Request: The client sends a request to accept the offered IP address.
4. Acknowledge: The DHCP server confirms the request and assigns the IP address to the client.

Result: The client successfully receives an IP address automatically from the DHCP server and can communicate over the network.

3. **Change the DHCP pool range on your Cisco Packet Tracer DHCP server so that only one IP address is available. Add two PCs and observe what happens when both try to get an IP address. Take a screenshot of the result and explain in 2-3 lines why only one PC succeeds.**

➢



DHCP Pool Limitation Experiment

Procedure:

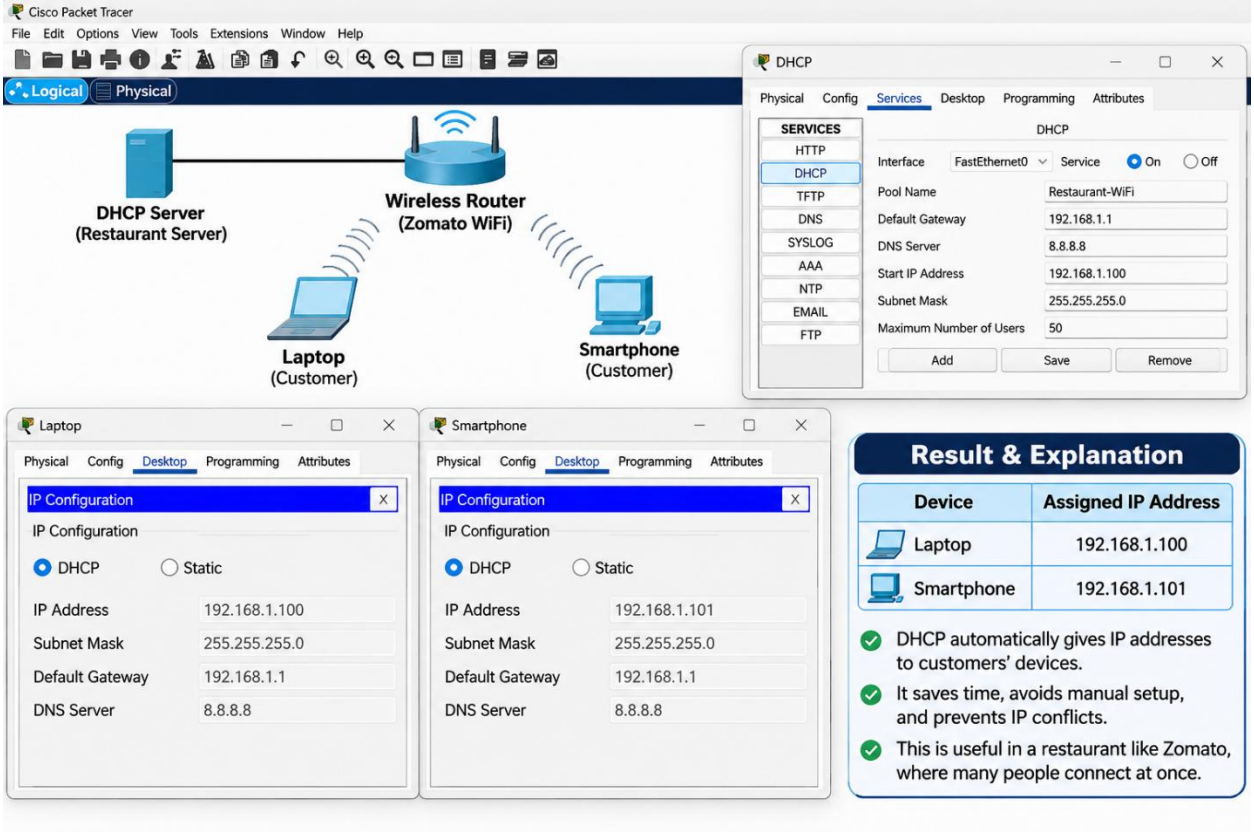
1. Configured the DHCP server with a single available IP address by setting Maximum Number of Users to 1.
2. Connected two PCs to the network switch.
3. Selected DHCP in the IP Configuration window of both PCs.
4. Observed the IP address assignment on both PCs.

Observation:

- PC1: Successfully received IP address 192.168.1.2.
- PC2: Failed to receive an IP address because no available addresses remained in the DHCP pool.

4. **Simulate a Zomato-style restaurant WiFi scenario in Cisco Packet Tracer: configure a DHCP server to provide IP addresses to both a laptop and a smartphone (use a generic PC as smartphone). Show the assigned IPs and explain why DHCP is useful in this situation.**

➢



Zomato-Style Restaurant Wi-Fi Network Using DHCP

Procedure:

1. Created a restaurant Wi-Fi network in Cisco Packet Tracer using a wireless router, one DHCP server, one laptop, and one generic PC representing a smartphone.
2. Configured the DHCP server and enabled the DHCP service.
3. Created a DHCP pool named Restaurant-WiFi.
4. Set the starting IP address to 192.168.1.100 and subnet mask to 255.255.255.0.
5. Connected both devices to the Wi-Fi network and selected DHCP in their IP Configuration settings.

Assigned IP Addresses:

Device	Assigned IP Address
Laptop	192.168.1.100
Smartphone (Generic PC)	192.168.1.101

Why is DHCP Useful?

DHCP automatically assigns IP addresses to customers' devices when they connect to restaurant Wi-Fi. It saves time, avoids manual IP configuration, prevents IP address conflicts, and allows multiple customers to connect easily.

Result:

Both the laptop and smartphone successfully received different IP addresses automatically from the DHCP server. This demonstrates how DHCP helps manage customer connections in a restaurant Wi-Fi network like Zomato.